THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET 1

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{25}{9}$	$1+\tan^2\theta$
2	**(d)** $2+2\sqrt{3}$ m	$h(\cot 30^\circ-\cot 45^\circ)=4$ gives $h=4\div(\sqrt{3}-1)$; Height $=2+2\sqrt{3}$ m (distances $6+2\sqrt{3}$ m and $2+2\sqrt{3}$ m)
3	**(b)** 48 cm	$AM=\frac{30\times40}{50}=24$ cm; $AB=2AM=48$ cm
4	**(a)** 4.47 cm²	Sector $=25.67$ cm^2; triangle $=\frac{\sqrt{3}}{4}r^2=21.19$ cm^2; Segment $=25.67-21.19=4.47$ cm^2
5	**(a)** $-\frac{7}{25}$	$\cos^2\theta-\sin^2\theta$
6	**(c)** 20 m	Height of the top of the flagstaff $=d\tan 60^\circ=30$ m; Length $=30-10=20$ m
7	**(c)** 14 cm	Perimeter $=PC+CE+ED+DP=PC+CA+DB+DP=PA+PB$; Perimeter $=2\times7=14$ cm
8	**(d)** 150 mm	Circumference $=\pi d=66$ mm; diameters $=4\times21=84$ mm; Total $=150$ mm
9	**(b)** $\frac{24}{25}$	$2\cdot\frac{p}{c}\cdot\frac{q}{c}$
10	**(c)** 2 minutes	Distance covered in 4 min $=h\sqrt{3}-\frac{h}{\sqrt{3}}=\frac{2h}{\sqrt{3}}$; distance left $=\frac{h}{\sqrt{3}}$, which is half of it; Time left $=4\div2=2$ minutes
11	**(b)** 49 cm	$15x+4=8x+25$ gives $x=3$; $QA=15(3)+4=49$ cm
12	**(a)** 1925 m²	Area $=\frac{180}{360}\times\frac{22}{7}\times35^2$; Area $=1925$ m^2
13	**(c)** $\frac{1}{3}$	$(\sec+\tan)(\sec-\tan)=1$
14	**(a)** 12 m	Both objects are on the same side, so the distances are subtracted; Distance $=12$ m
15	**(a)** 120 cm	Half-chord $=\sqrt{68^2-32^2}=60$ cm; Chord $=2\times60=120$ cm

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$\cos^2\theta=1-\left(\frac{4}{5}\right)^2=\frac{9}{25}$ (1); value $=-\frac{7}{25}$ (1).	1+1
17	Let height $=h$; the distances from the foot are $h\cot45^\circ$ and $h\cot60^\circ$, which differ by 6 m; $h(\cot45^\circ-\cot60^\circ)=6$ gives $h=6\div(1-\frac{\sqrt{3}}{3})$ (1); Height $=9+3\sqrt{3}$ m (distances $9+3\sqrt{3}$ m and $3+3\sqrt{3}$ m) (1).	1+1
18	Area of $\triangle OAP=\frac{1}{2}\times60\times45$ and also $=\frac{1}{2}\times OP\times AM$ where $AM\perp OP$ (M is the midpoint of AB); $AM=\frac{60\times45}{75}=36$ cm (1); $AB=2AM=72$ cm (1).	1+1
19	The 14 ribs divide the circle into 14 equal sectors; Area of the circle $=\frac{22}{7}\times49^2=7546$ cm^2 (1); Area between ribs $=7546\div14=539$ cm^2 (1).	1+1
20	Hypotenuse $=29$ (1); $\sin\theta=\frac{21}{29}$, $\sec\theta=\frac{29}{20}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	LHS $=\frac{\sin^2\theta+\cos^2\theta}{\sin\theta\cos\theta}$ (2); $=\frac{1}{\sin\theta\cos\theta}=\sec\theta\csc\theta$ (1).	2+1
22	Let the tower be H and the distance d: $H=d\tan60^\circ$ and $H-18\sqrt{3}=d\tan30^\circ$ (1); Subtract: $d(\tan60^\circ-\tan30^\circ)=18\sqrt{3}$, so $d=27$ (1); $H=d\tan60^\circ=27\sqrt{3}$; tower $27\sqrt{3}$ m, distance 27 m (1).	1+1+1
23	Area of $\triangle OAP=\frac{1}{2}\times30\times40$ and also $=\frac{1}{2}\times OP\times AM$ where $AM\perp OP$ (M is the midpoint of AB) (1); $AM=\frac{30\times40}{50}=24$ cm (1); $AB=2AM=48$ cm (1).	1+1+1
24	Area of sector $=\frac{\theta}{360}\times\pi r^2$ (1); $\frac{50}{360}\times\frac{22}{7}\times42^2$ (1); Area $=770$ cm^2 (1).	1+1+1
	OR: Square both sides: $1+2\sin\theta\cos\theta=\frac{49}{25}$ (2); $\sin\theta\cos\theta=\frac{12}{25}$ (1).	2+1
25	Distance of the point from the building: $d=14\cot45^\circ=14$ m (1); Height of the top of the flagstaff $=d\tan60^\circ=14\sqrt{3}$ m (1); Length $=14\sqrt{3}-14=14\sqrt{3}-14$ m (1).	1+1+1

	OR: $AC = 85$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 36 - r$, $CD = BC - r = 77 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(36 - r) + (77 - r) = 85$ gives $r = 14$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$\csc\theta - \sin\theta = \dfrac{\cos^2\theta}{\sin\theta}$, $\sec\theta - \cos\theta = \dfrac{\sin^2\theta}{\cos\theta}$ (2); $\tan\theta + \cot\theta = \dfrac{1}{\sin\theta\cos\theta}$ (1); product $= \sin\theta\cos\theta \cdot \dfrac{1}{\sin\theta\cos\theta} = 1$ (2).	5
	OR: Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 4 = d\tan 45^\circ$ (2); Subtract: $d(\tan 60^\circ - \tan 45^\circ) = 4$, so $d = 2 + 2\sqrt{3}$ (2); $H = d\tan 60^\circ = 6 + 2\sqrt{3}$; tower $6 + 2\sqrt{3}$ m, distance $2 + 2\sqrt{3}$ m (1).	2+2+1
27	$XA = XC$ (tangents from X) and $YB = YC$ (tangents from Y) (2); $XY = XC + CY$ since C lies on XY, so $XY = AX + BY$ (1); Perimeter of $\triangle PXY = PX + XY + YP = PX + XA + BY + YP$ (1); $= PA + PB = 2PA$ because $PA = PB$ (1).	2+1+1+1
	OR: Segment = sector − triangle (2); Sector $= 641.67$ cm^2; triangle $= \frac{\sqrt{3}}{4}r^2 = 529.81$ cm^2 (2); Segment $= 641.67 - 529.81 = 111.85$ cm^2 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 117 cm	Arc $= \frac{45}{360} \times 2\pi \times 42 = 33$ cm; Perimeter $= 33 + 84 = 117$ cm
2	**(a)** 1	Each pair sums to 1
3	**(d)** 4 m	Part of the tower above the top of the building $= d\tan 30^\circ = 1$ m; Tower $= 4$ m
4	**(b)** 15 cm	$AF = AB - r = 45 - r$, $CD = BC - r = 60 - r$ and $AF + CD = AC$ (tangent lengths); $(45 - r) + (60 - r) = 75$ gives $r = 15$ cm
5	**(b)** 30°	$\theta = \frac{22 \times 360}{2 \times \frac{22}{7} \times 42}$; $\theta = 30^\circ$
6	**(d)** 1	Pair as $\tan\theta \tan(90^\circ - \theta) = 1$
7	**(c)** $9 + 3\sqrt{3}$ m	The points are on opposite sides, so the distances add; Distance $= 9 + 3\sqrt{3}$ m
8	**(d)** 36 cm	$PT^2 = OP^2 - OT^2 = 85^2 - 77^2 = 1296$; $PT = 36$ cm
9	**(b)** 1400 cm²	Circle $= 3850$ cm²; square $= 2450$ cm²; Difference $= 1400$ cm²
10	**(d)** $\frac{169}{25}$	$1 + \tan^2\theta$
11	**(d)** 30°	$\tan\theta = \frac{\sqrt{3}}{3}$ which is $\tan 30^\circ$; $\theta = 30^\circ$
12	**(c)** the perpendicular bisector of AB	$PA = PB$ and $OA = OB$
13	**(a)** 924 m²	Increase $= \frac{1}{4} \times \frac{22}{7}(35^2 - 7^2)$; Increase $= 924$ m²
14	**(a)** $-\frac{119}{169}$	$\cos^2\theta - \sin^2\theta$
15	**(a)** $3 + 3\sqrt{3}$ m	$h(\cot 30^\circ - \cot 45^\circ) = 6$ gives $h = 6 \div (\sqrt{3} - 1)$; Height $= 3 + 3\sqrt{3}$ m (distances $9 + 3\sqrt{3}$ m and $3 + 3\sqrt{3}$ m)

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Distance of the point from the building: $d = 9\cot 45^\circ = 9$ m; Height of the top of the flagstaff $= d\tan 30^\circ = 3\sqrt{3}$ m (1); Length $= 3\sqrt{3} - 9 = 3\sqrt{3} - 9$ m (1).	1+1
17	$CA = CE$ and $DB = DE$ (tangents from C and D); Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB$ (1); Perimeter $= 2 \times 9 = 18$ cm (1).	1+1
18	Circumference $= 2 \times \frac{22}{7} \times 35 = 220$ cm; Distance 110 m $= 11000$ cm; revolutions $= \frac{11000}{220}$ (1); Revolutions $= 50$ (1).	1+1
19	$\frac{x^2}{4} = \sec^2\theta$, $\frac{y^2}{9} = \tan^2\theta$ (1); difference $= 1$ (1).	1+1
20	Let height $= h$: distances $h\cot 30^\circ = h\sqrt{3}$ and $h\cot 60^\circ = \frac{h}{\sqrt{3}}$; Distance covered in 6 min $= h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$; distance left $= \frac{h}{\sqrt{3}}$, which is half of it (1); Time left $= 6 \div 2 = 3$ minutes (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	The quadrant has radius s, so its area is $\frac{1}{4}\pi s^2$ (1); Quadrant $= 3118.5$ cm²; square $= 3969$ cm² (1); Remaining $= 850.5$ cm² (1).	1+1+1
22	Square: $\tan^2\theta + 2 + \cot^2\theta = 9$ (2); value $= 7$ (1).	2+1
23	Distances of the points from the foot: $h\cot 45^\circ = 6$ m and $h\cot 60^\circ = 2\sqrt{3}$ m (1); The points are on opposite sides, so the distances add (1); Distance $= 6 + 2\sqrt{3}$ m (1).	1+1+1
24	Tangents from an external point are equal: $QA = QB$ (1); $21x + 5 = 13x + 29$ gives $x = 3$ (1); $QA = 21(3) + 5 = 68$ cm (1).	1+1+1
	OR: In 5 minutes the hand turns through $6^\circ \times 5 = 30^\circ$ (1); Area $= \frac{30}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 462$ cm² (1).	1+1+1
25	LHS $= \frac{1}{\cos^2\theta} + \frac{1}{\sin^2\theta} = \frac{\sin^2\theta + \cos^2\theta}{\sin^2\theta\cos^2\theta}$ (2); $= \sec^2\theta\csc^2\theta$ (1).	2+1

	OR: Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 20\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 20\sqrt{3}$, so $d = 30$ (1); $H = d\tan 60^\circ = 30\sqrt{3}$; tower $30\sqrt{3}$ m, distance 30 m (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Divide numerator and denominator by $\cos A$ (1): $\dfrac{5\tan A + 4}{7\tan A + 3}$ (1); $= \dfrac{5(\frac{24}{7}) + 4}{7(\frac{24}{7}) + 3} = \frac{148}{189}$ (2); verified with $\sin A = \frac{24}{25}$, $\cos A = \frac{7}{25}$ (1).	1+1+2+1
	OR: Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 48 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 48$ gives $h = 12\sqrt{3}$ (2); Height $12\sqrt{3}$ m; distances 12 m and 36 m (1).	2+2+1
27	Area of $\triangle OAP = \frac{1}{2} \times 72 \times 21$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB) (2); $AM = \frac{72\times 21}{75} = 20.16$ cm (2); $AB = 2AM = 40.32$ cm (1).	2+2+1
	OR: Required area $= \frac{\theta}{360} \times \pi(R^2 - r^2)$ (2); $\frac{36}{360} \times \frac{22}{7} \times (21^2 - 14^2)$ (2); Area $= 77$ cm^2 (1).	2+2+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 26.88 cm	$AM = \frac{14\times48}{50} = 13.44$ cm; $AB = 2AM = 26.88$ cm
2	**(c)** 67.82 cm²	Sector $= 115.50$ cm^2; triangle with sides r, r and angle 120° has area $\frac{1}{2}r^2\sin120^\circ = \frac{\sqrt{3}}{4}r^2 = 47.68$ cm^2; Segment $= 115.50 - 47.68 = 67.82$ cm^2
3	**(d)** $\frac{120}{169}$	$2 \cdot \frac{p}{c} \cdot \frac{q}{c}$
4	**(a)** 24 m	Height of the top of the flagstaff $= d\tan60^\circ = 36$ m; Length $= 36 - 12 = 24$ m
5	**(a)** 22 cm	Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB$; Perimeter $= 2 \times 11 = 22$ cm
6	**(a)** 497 mm	Circumference $= \pi d = 154$ mm; diameters $= 7 \times 49 = 343$ mm; Total $= 497$ mm
7	**(b)** $\frac{1}{5}$	$(\sec+\tan)(\sec-\tan) = 1$
8	**(b)** 4 minutes	Distance covered in 8 min $= h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$; distance left $= \frac{h}{\sqrt{3}}$, which is half of it; Time left $= 8 \div 2 = 4$ minutes
9	**(b)** 43 cm	$12x + 7 = 5x + 28$ gives $x = 3$; $QA = 12(3) + 7 = 43$ cm
10	**(a)** 2772 m²	Area $= \frac{180}{360} \times \frac{22}{7} \times 42^2$; Area $= 2772$ m^2
11	**(b)** 1	Each pair sums to 1
12	**(d)** 18 m	Both objects are on the same side, so the distances are subtracted; Distance $= 18$ m
13	**(b)** 112 cm	Half-chord $= \sqrt{70^2 - 42^2} = 56$ cm; Chord $= 2 \times 56 = 112$ cm
14	**(d)** 139 cm	Arc $= \frac{75}{360} \times 2\pi \times 42 = 55$ cm; Perimeter $= 55 + 84 = 139$ cm
15	**(d)** 1	Pair as $\tan\theta\tan(90^\circ - \theta) = 1$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord; Half-chord $= \sqrt{90^2 - 72^2} = 54$ cm (1); Chord $= 2 \times 54 = 108$ cm (1).	1+1
17	Outer radius $= 38.5$ m; Area $= \pi(R^2 - r^2) = \frac{22}{7}(38.5^2 - 35^2)$ (1); Area of path $= 808.5$ m^2 (1).	1+1
18	$\cos^2\theta = 1 - \left(\frac{12}{13}\right)^2 = \frac{25}{169}$ (1); value $= -\frac{119}{169}$ (1).	1+1
19	Let height $= h$; the distances from the foot are $h\cot45^\circ$ and $h\cot60^\circ$, which differ by 12 m; $h(\cot45^\circ - \cot60^\circ) = 12$ gives $h = 12 \div (1 - \frac{\sqrt{3}}{3})$ (1); Height $= 18 + 6\sqrt{3}$ m (distances $18 + 6\sqrt{3}$ m and $6 + 6\sqrt{3}$ m) (1).	1+1
20	Area of $\triangle OAP = \frac{1}{2} \times 72 \times 21$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB); $AM = \frac{72\times21}{75} = 20.16$ cm (1); $AB = 2AM = 40.32$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Area of $\triangle OAP = \frac{1}{2} \times 14 \times 48$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB) (1); $AM = \frac{14\times48}{50} = 13.44$ cm (1); $AB = 2AM = 26.88$ cm (1).	1+1+1
22	Area of sector $= \frac{\theta}{360} \times \pi r^2$ (1); $\frac{80}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 1232$ cm^2 (1).	1+1+1
23	Square both sides: $1 + 2\sin\theta\cos\theta = \frac{169}{100}$ (2); $\sin\theta\cos\theta = \frac{69}{200}$ (1).	2+1
24	Distance of the point from the building: $d = 16\cot45^\circ = 16$ m (1); Height of the top of the flagstaff $= d\tan60^\circ = 16\sqrt{3}$ m (1); Length $= 16\sqrt{3} - 16 = 16\sqrt{3} - 16$ m (1).	1+1+1
	OR: $AC = 85$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 40 - r$, $CD = BC - r = 75 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(40 - r) + (75 - r) = 85$ gives $r = 15$ cm (1).	1+1+1
25	The quadrant has radius s, so its area is $\frac{1}{4}\pi s^2$ (1); Quadrant $= 3850$ cm^2; square $= 4900$ cm^2 (1); Remaining $= 1050$ cm^2 (1).	1+1+1

	OR: Square: $\tan^2\theta + 2 + \cot^2\theta = 16$ (2); value $= 14$ (1).	2+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	LHS $= \dfrac{\sin\theta(1 - 2\sin^2\theta)}{\cos\theta(2\cos^2\theta - 1)}$ (2); $1 - 2\sin^2\theta = 2\cos^2\theta - 1$ (2); so LHS $= \tan\theta$ (1).	5
	OR: Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 6 = d\tan 45^\circ$ (2); Subtract: $d(\tan 60^\circ - \tan 45^\circ) = 6$, so $d = 3 + 3\sqrt{3}$ (2); $H = d\tan 60^\circ = 9 + 3\sqrt{3}$; tower $9 + 3\sqrt{3}$ m, distance $3 + 3\sqrt{3}$ m (1).	2+2+1
27	Let the incircle touch AB, BC, CA at F, D, E and let the centre be O; $OF \perp AB$, $OD \perp BC$ (1); $BFOD$ has three right angles and $OF = OD = r$, so it is a square: $BF = BD = r$ (2); $AE = AF = AB - r$ and $CE = CD = BC - r$ (equal tangents) (1); $AC = AE + EC = AB + BC - 2r$, so $r = \frac{AB + BC - AC}{2}$ (1).	1+2+1+1
	OR: Segment $=$ sector $-$ triangle (2); Sector $= 1848.00$ cm^2; triangle with sides r, r and angle 120° has area $\frac{1}{2}r^2\sin 120^\circ = \frac{\sqrt{3}}{4}r^2 = 762.93$ cm^2 (2); Segment $= 1848.00 - 762.93 = 1085.07$ cm^2 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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